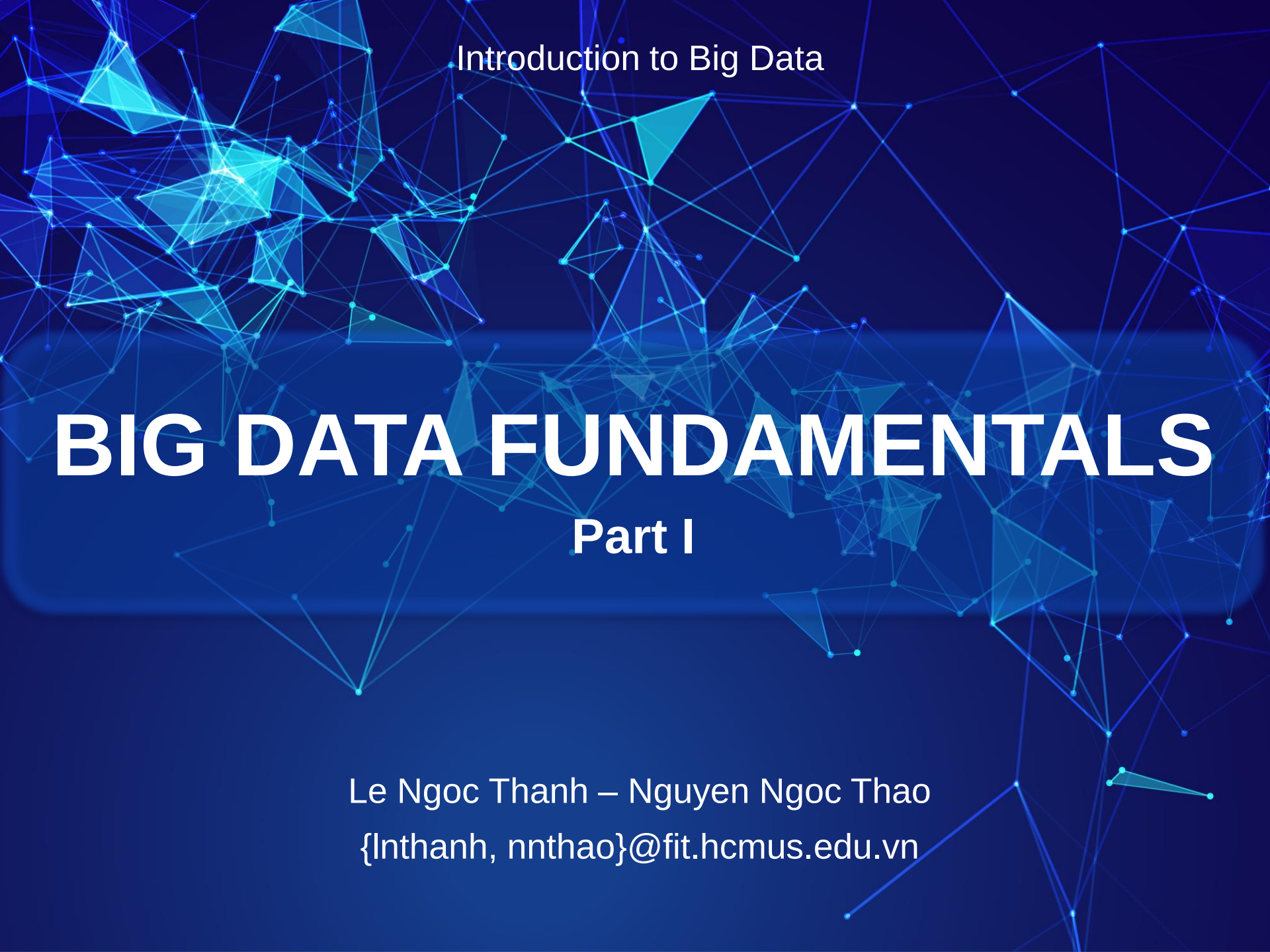




Introduction to Big Data

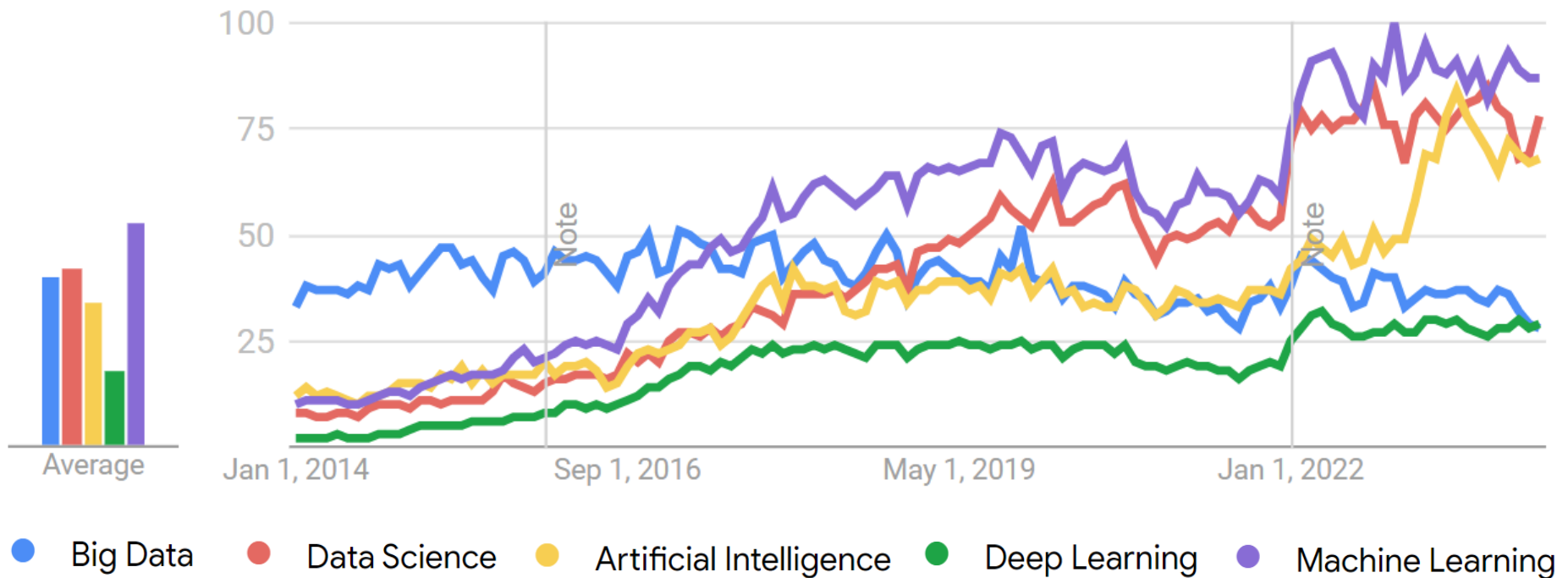
BIG DATA FUNDAMENTALS

Part I

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Big data: The trending term

- **Big data** is among trending search terms in recent years.



Source: Google Trends, updated 01/2024

Outline

- What is Big data?
 - The definitions of Big data
 - The V's characteristics of Big data
 - Common Issues in Big data
- Big data case studies
 - The applications of Big data
 - Big data projects in practice
- Motivations and opportunities

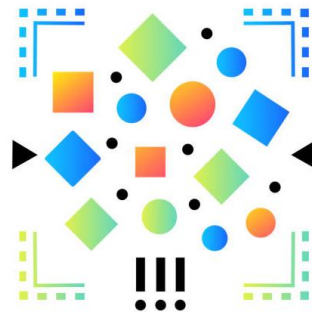
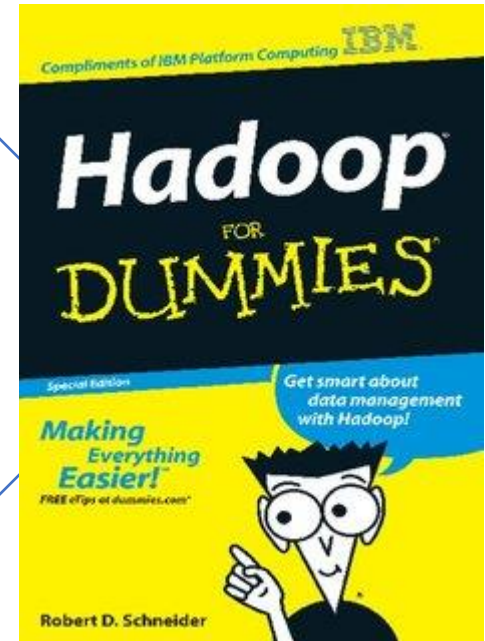
What is Big data?

It is not big. It is just bigger...

Big data: A definition

Big data is a term that describes at least three separate, but interrelated, trends.

- ✓ Capturing and managing lots of information
- ✓ Working with many new types of data
- ✓ Exploiting these masses of information and new data types with new styles of applications



Big data: A definition

- A variety of definitions for Big data are available worldwide.

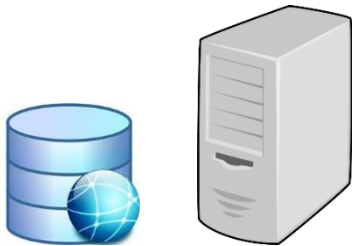
Big data is a term used to refer to the *study and applications of data sets that are so big and complex* that traditional data-processing application software are inadequate to deal with them. – Wikipedia.

Big data is *high-volume, high-velocity and/or high-variety information assets* that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation. – Garner, 2001.

Big data refers to *the dynamic, large and disparate volumes of data being created by people, tools and machines*; it requires new, innovative and scalable technology to collect, host and analytically process the vast amount of data gathered in order to derive real-time business insights that relate to consumers, risk, profit, performance, productivity management and enhanced shareholder value. – Ernst & Young, 2014.

Small data vs. Big data

- You may treat “Big data” as “small data” that grows bigger.
- The new scale of data may require novel approaches for techniques and frameworks.



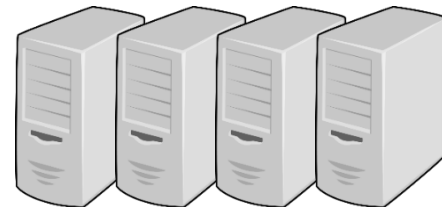
small data, small computer



bigger data



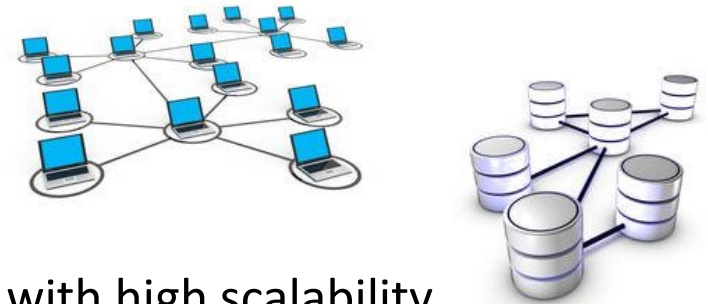
bigger
computer?



or more small computers?

Technologies in Big data

- Big data collects **both old and new technologies** to **gain from data actionable insights**.



There is an urgent need for **parallel processing** in **distributed environment** with high scalability.

- We are now able **solve new problems or existing problems in a better way**.
- A huge volume of disparate data can be managed, at the right speed, and within the right time frame
→ allow for **real-time analysis and reaction**

The 5V's characteristics

- The characteristics of Big data are characterized by the 5V's.



VOLUME



VELOCITY



VARIETY



VALUE



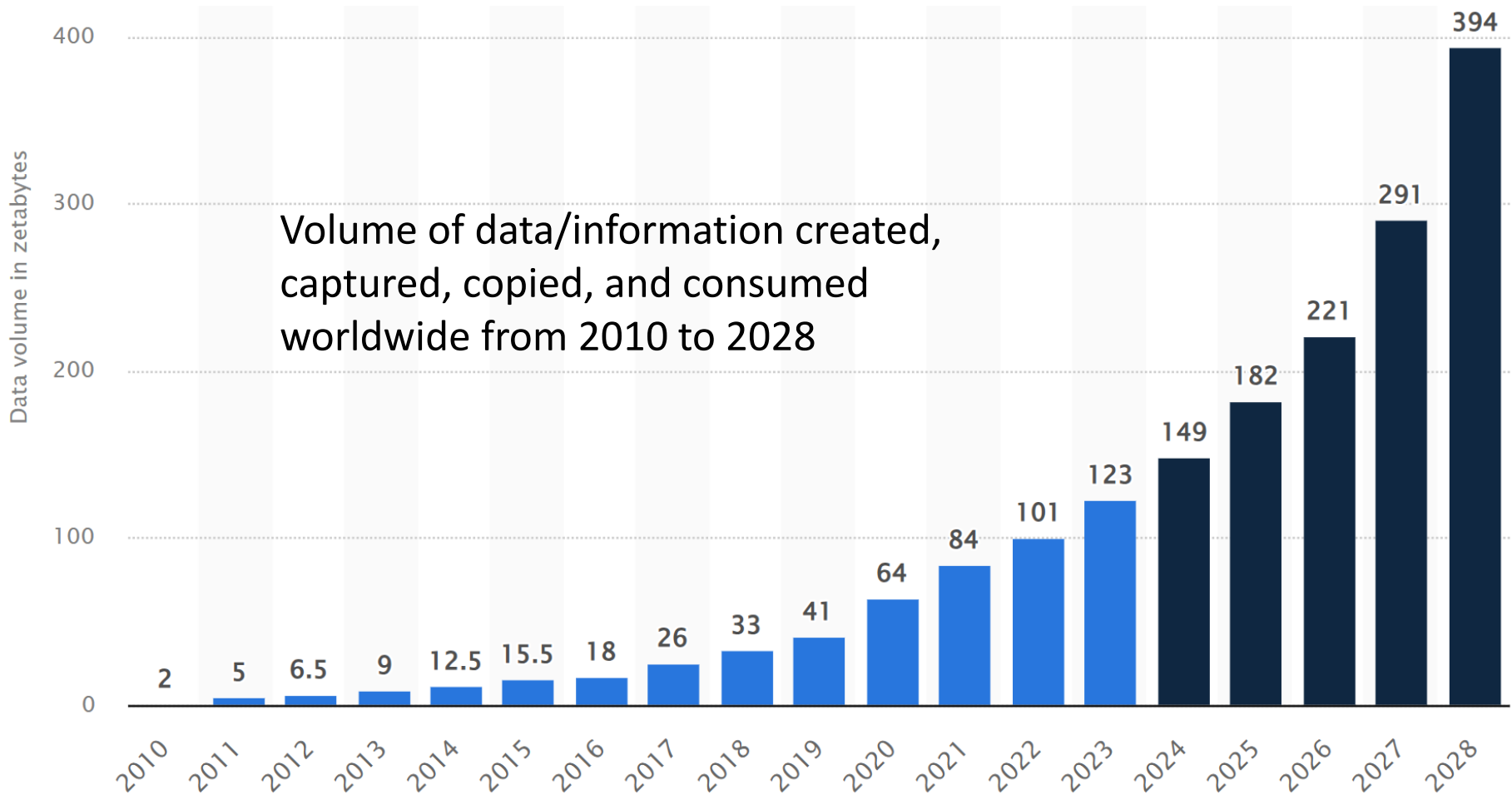
VERACITY

The 5V's characteristics

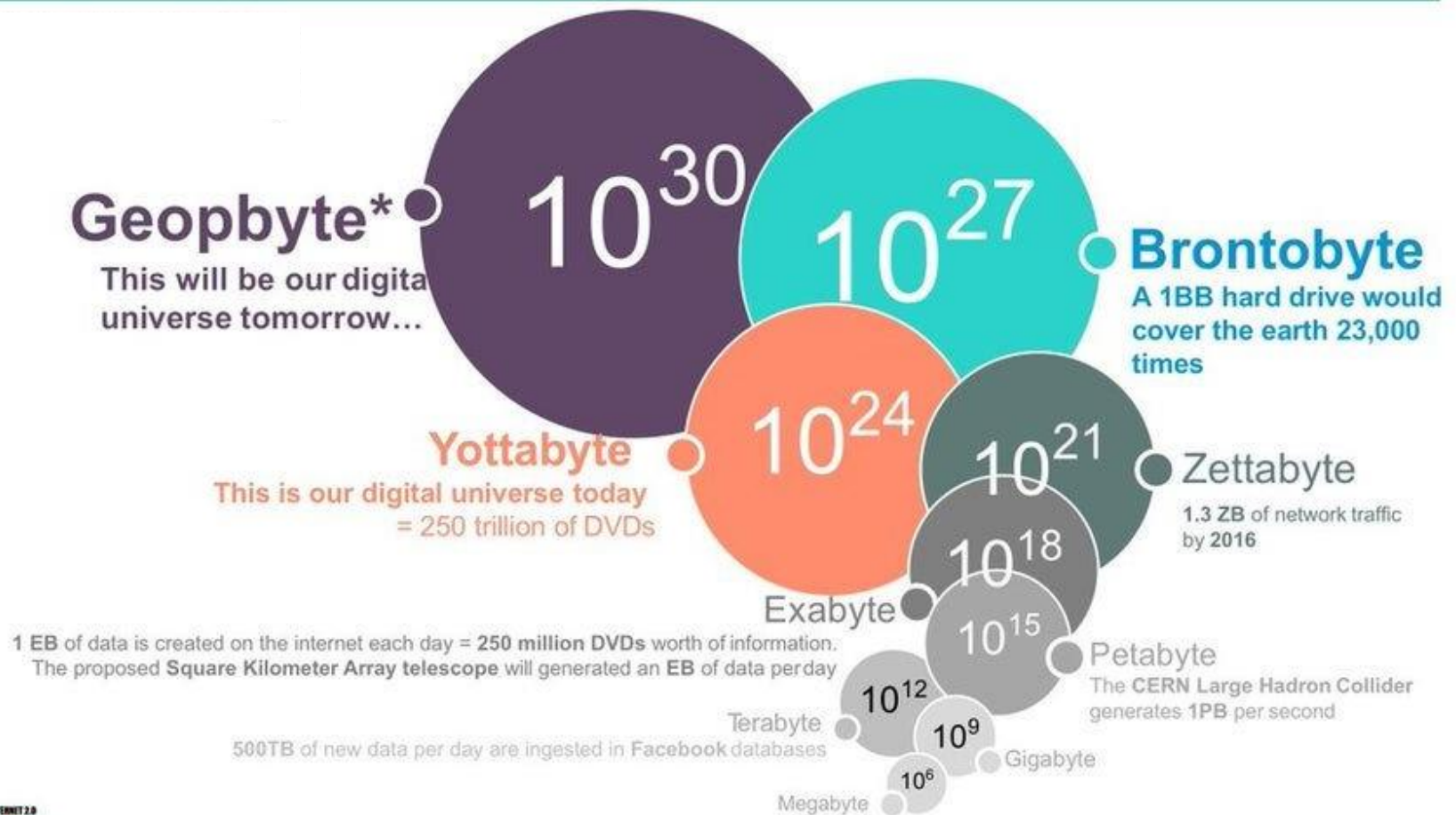


- **Description:** The amount of data generated is vast compared to traditional data sources.
- **Attributes:** Exabyte, zettabyte, yottabytes, etc.
- **Drivers:** Increase in data sources, higher resolution sensors, scalable infrastructure.

The explosive growth of data



New data units for the Big data era



3 Important Statistics About How Much Data Is Created Every Day

1 How much data is generated every minute?

Source: Domo

 **41,666,667**

messages shared
by WhatsApp users

 **1,388,889**

video / voice calls made
by people worldwide

 **404,444**

hours of video streamed
by Netflix users

 **347,222**

stories posted by Instagram users

 **150,000**

messages shared by Facebook users

 **147,000**

photos shared by Facebook users

2 Estimated Data Consumption from 2021 to 2024

Source: IDC / Statista



3 Data Growth in 2021

Sources: TechJury, Internet Live Stats, Cisco, PurpleSec

 **2 TRILLION**

searches on Google by the end of 2021

 **1.134 TRILLION MB**

volume of data created every day

 **3,026,626**

emails sent every second, 67% of which are spam

 **278,108 PETABYTES**

global IP data per month by the end of 2021

 **230,000**

new malware versions created every day

 **82%**

share of video in total global internet
traffic at the end of 2021

Why does data become big now?

- Key enablers of appearance and growth of data are



Increase of storage capacity



Increase of processing power



Availability of data

The 5V's characteristics



- **Description:** Data is being generated extremely fast, a process that never stops; and the speed at which data is transformed into insight
- **Attributes:** Batch; near/real-time; streams
- **Drivers:** Improved connectivity; competitive advantage; precomputed information

Real-time and/or fast data



Social media and networks
(all of us are generating data)



Scientific instruments
(collecting all sorts of data)



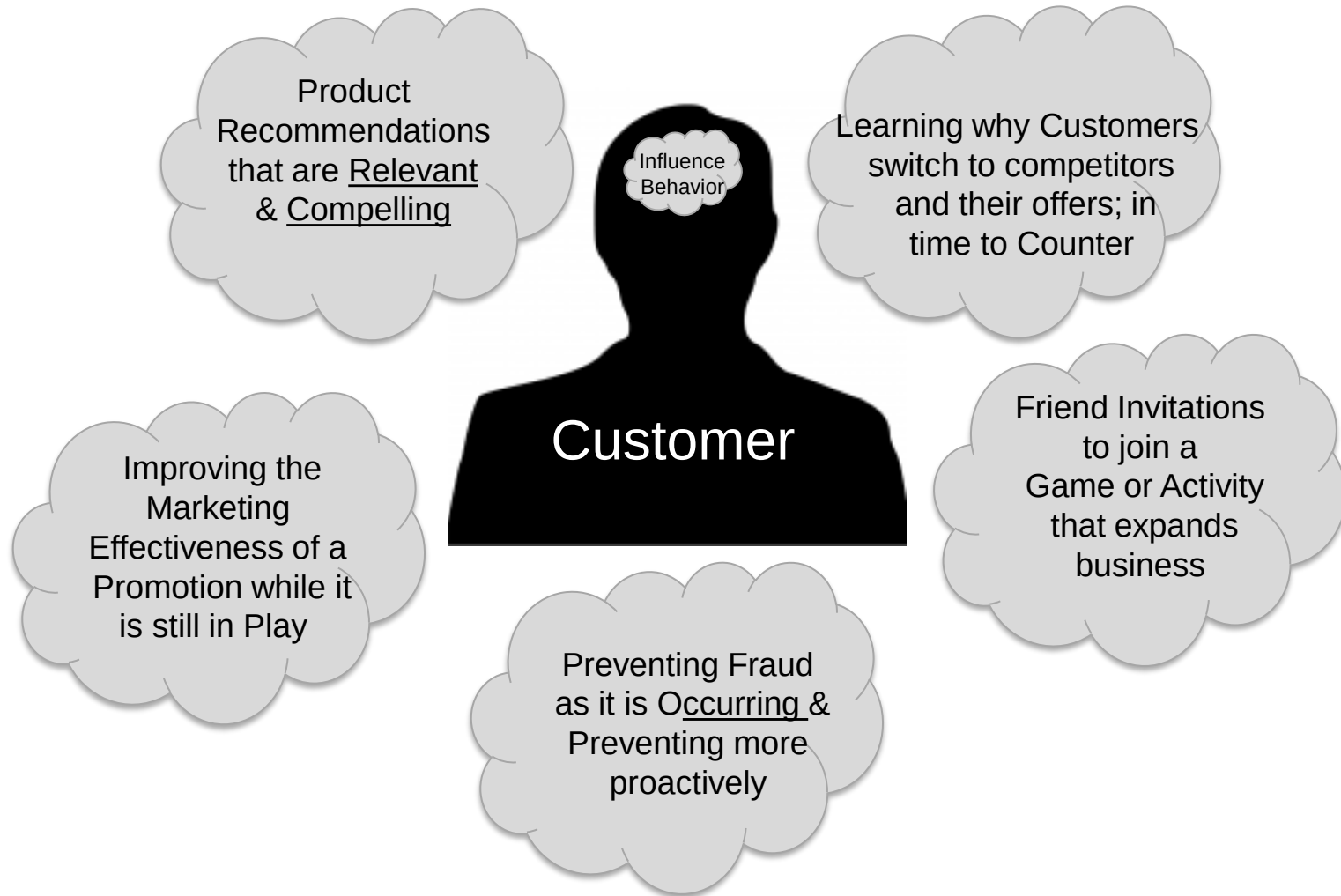
Mobile devices
(tracking all objects all the time)



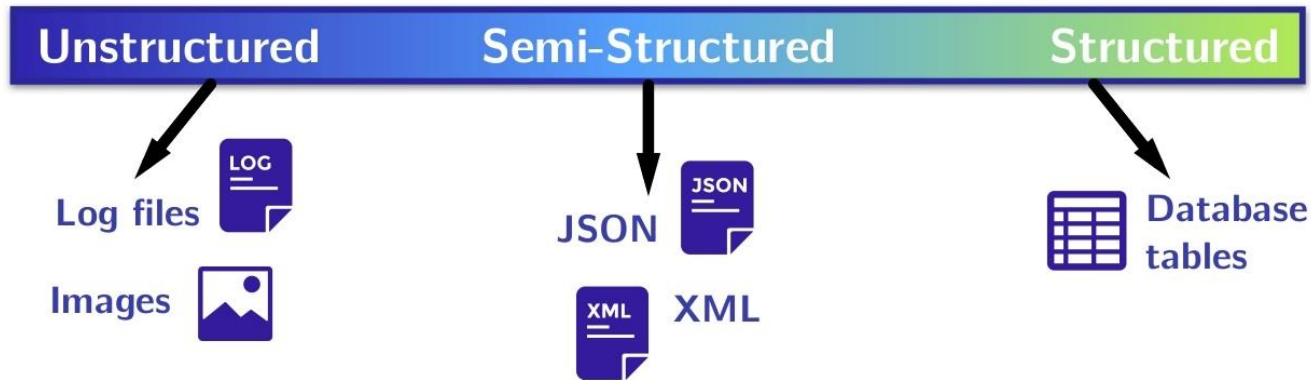
Sensor technology
and networks
(measuring all kinds of data)

- Innovations and their progresses are no longer hindered by [the ability to collect data](#) but by the [ability to manage, analyze, summarize, visualize, and discover knowledge from the collected data](#) in a timely manner and in a scalable fashion.

Real-time analytics/decision requirement

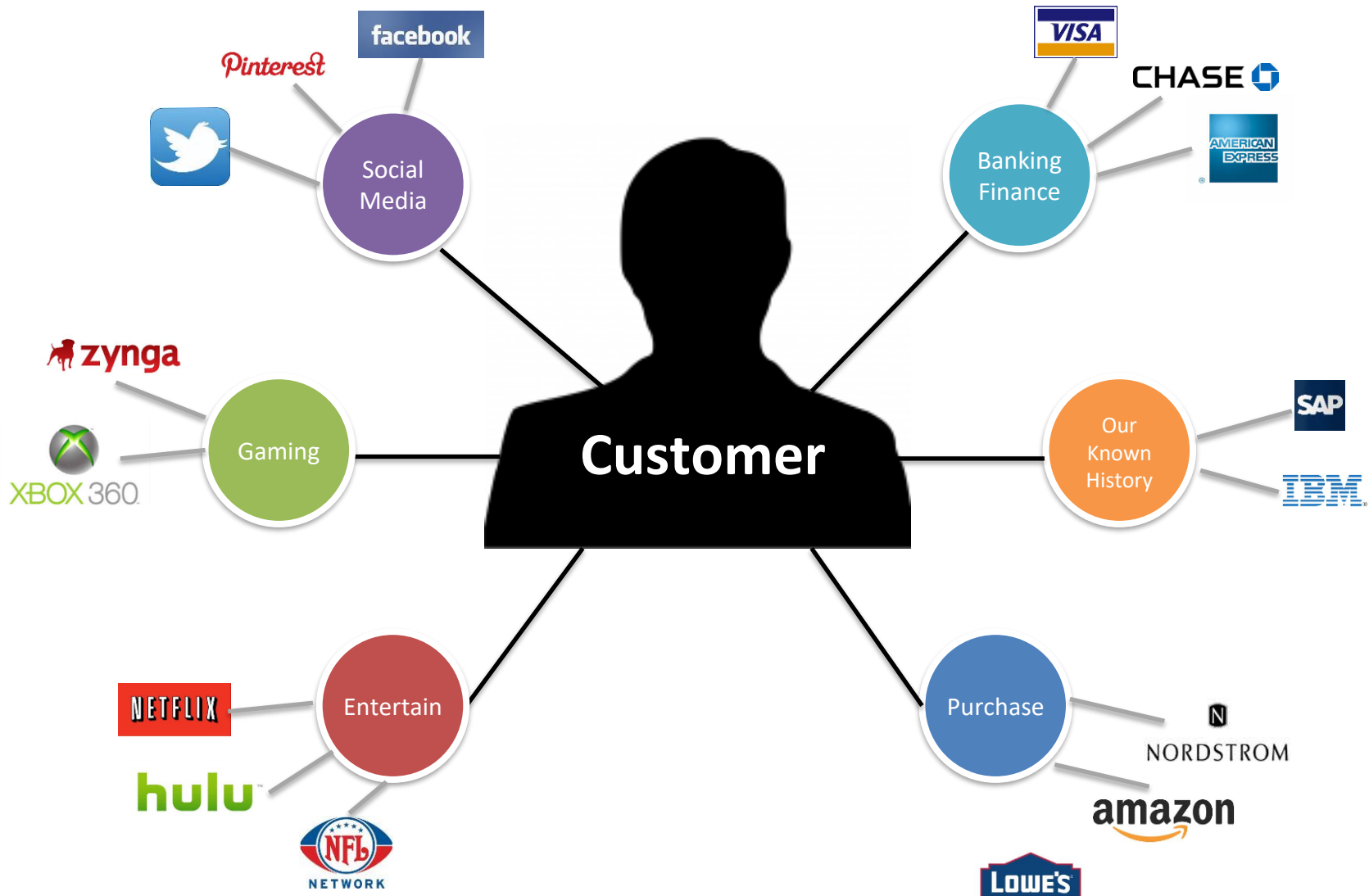


The 5V's characteristics



- **Description:** Data comes from different sources, machines, people, and processes both from outside and inside the organizations
- **Attributes:** Degree of structure; complexity
- **Drivers:** Mobile; social media; video; genomics; IoT

A single view to the customer



The 5V's characteristic



- **Description:** Quality and origin of data
- **Attributes:** Consistency; completeness; integrity; ambiguity
- **Drivers:** Cost; need of traceability and justification



- The ability and need to turn data into value
- Value is not only profit but also medical or social benefits, or personal satisfaction (customer, employee, etc.).

Common issues related to the 5V's

- As the data volume increases, the value of different data records will decrease in proportion to age, type, richness, and quantity among other factors.
- It is hard to handle complex data by existing traditional analytic systems.
 - Big data with relational databases, statistics/visualization packages
 - Massively parallel software running on tens, hundreds, or even thousands of computing units.
 - Data analytics with data that is constantly in motion.

Issues of personnel

- There is a considerable gap between Business leaders and IT professionals.



- Business leaders concern about adding value to their business and getting more and more profit.
- Meanwhile, IT leaders focus on the technicalities of the storage and processing only.

Issues of storage and transport

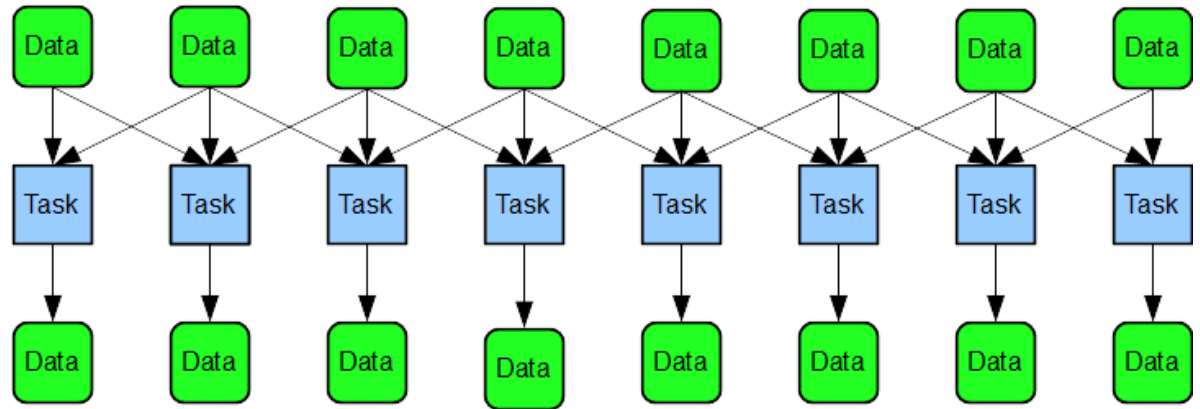
- Assume that we have an exabyte (10^{18}) of data.
- Modern hard disks can store several TBs (10^{12})
→ thousands of disks are required for an exabyte.
- A single computer system would be **unable to directly attach the requisite number of disks**.
- Accesses to that data also **overwhelms the networks**.
 - E.g., a 1GB/second network operating at 80% efficiency and a sustainable bandwidth of 100 MB/second would take approximately 14,465 days to transfer an exabyte of data.

Issues of data management

- Possibly the most difficult problem
- Issues of access, utilization, updating, governance, and reference (in publication) are major stumbling blocks.
- Data sources are varied by size, format, and by method of collection.
 - What, when, where, who, why and how it was collected.
- It is impractical to validate every data item in a huge source.

Issues of processing power

- Extensive parallel processing and new analytics algorithms are required.



Assume that an exabyte of data need to be processed and it is chunked into blocks of 8 words $\rightarrow$ 1 exabytes = 1K petabytes.

Assuming a processor expends 100 instructions on one block at 5 gigahertz $\rightarrow$ 1K petabytes would require a processing time of 635 years.



Big data case studies

The more data, the better decisions, and then the better outcomes...

Big data analytics: Applications



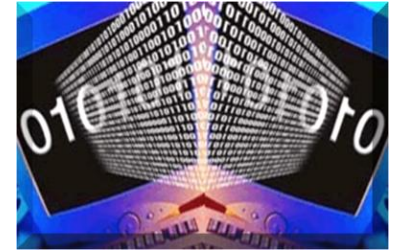
Smarter Healthcare



Multi-channel sales



Finance



Log Analysis



Homeland Security



Traffic Control



Telecom



Search Quality



Manufacturing



Trading Analytics

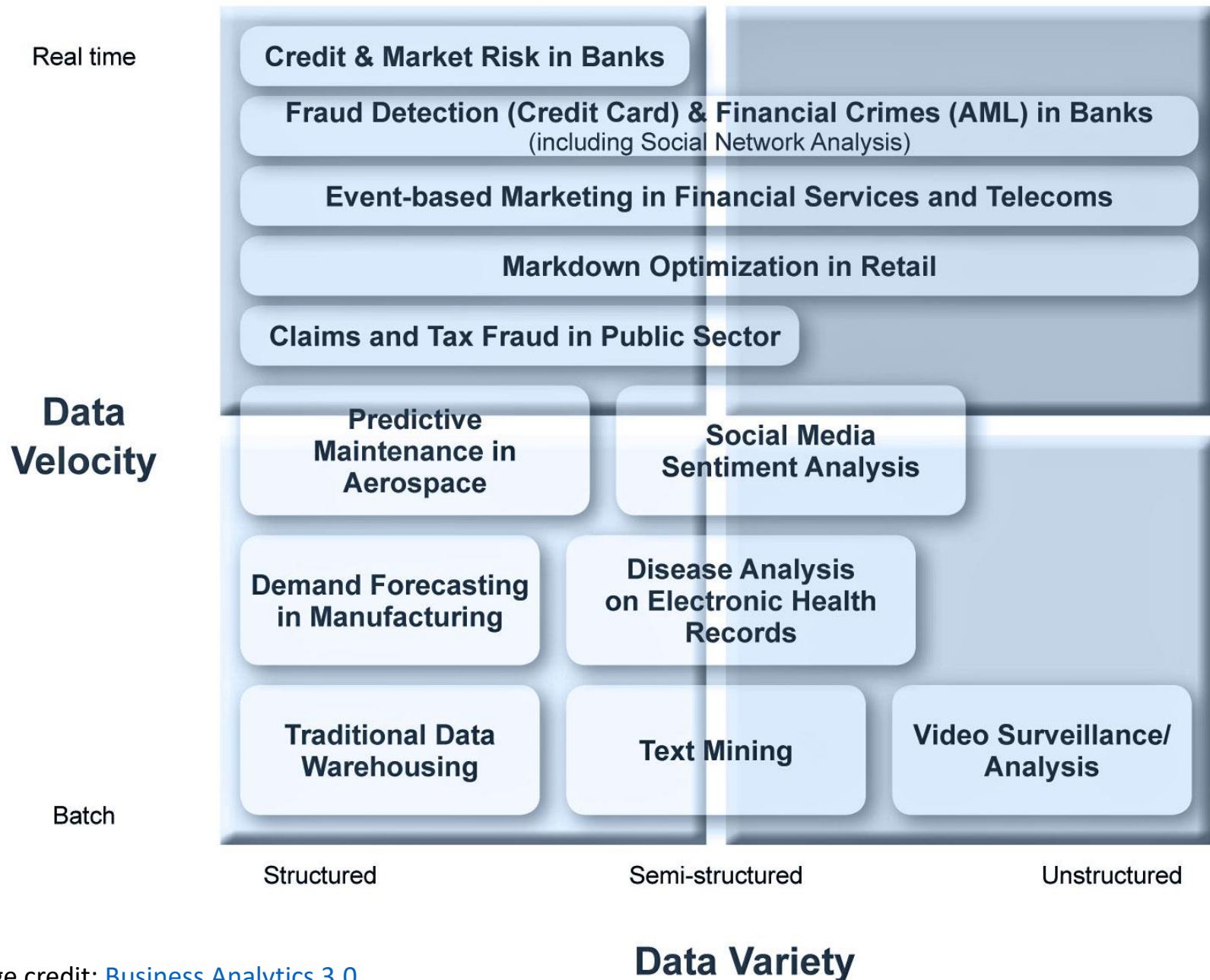


Fraud and Risk



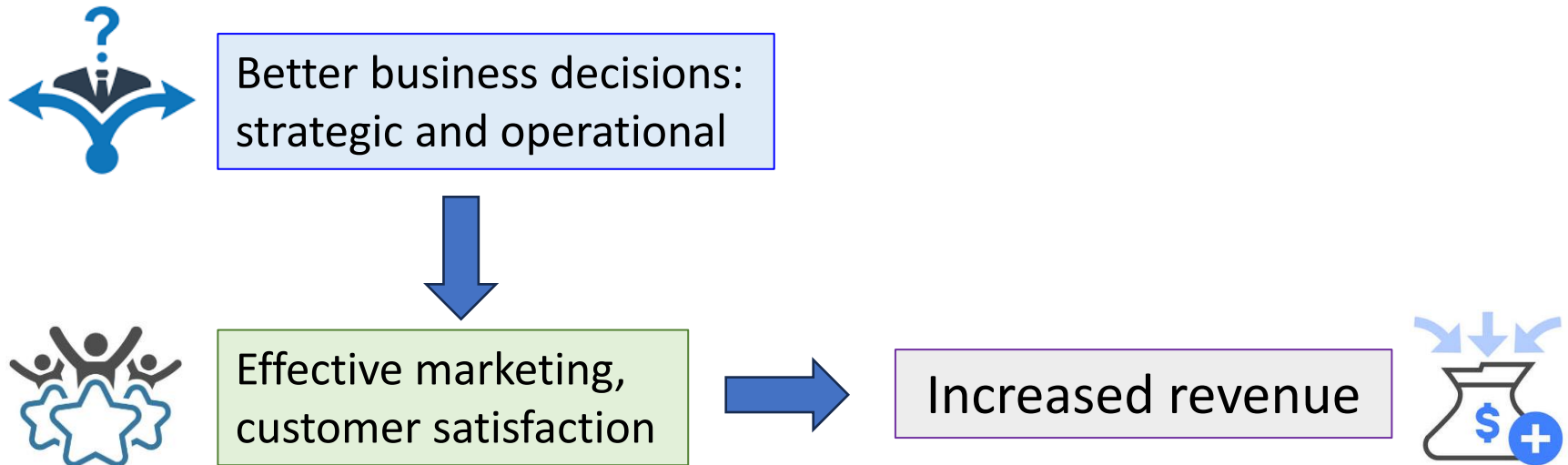
Retail: Churn, NBO

Potential Use Cases for Big Data Analytics



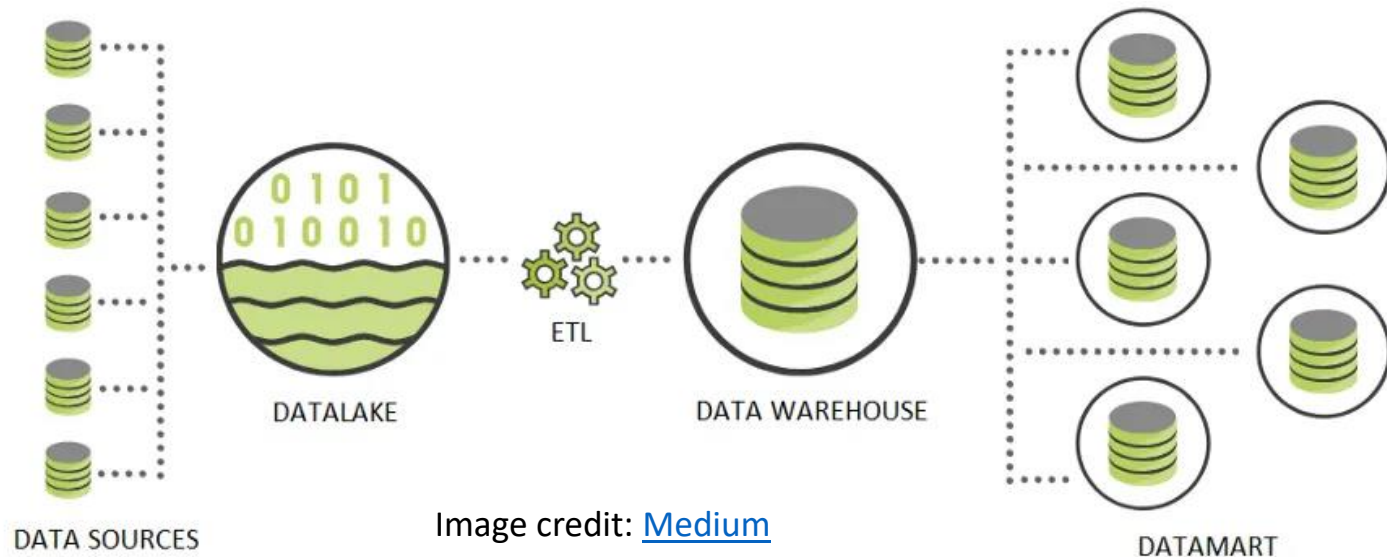
Big data analytics

- This analytics **exploits a large amount of data** for **interesting data relationships** to gain competitive advantage.
- Appropriate information: hidden patterns or correlations
- Competitive advantage:



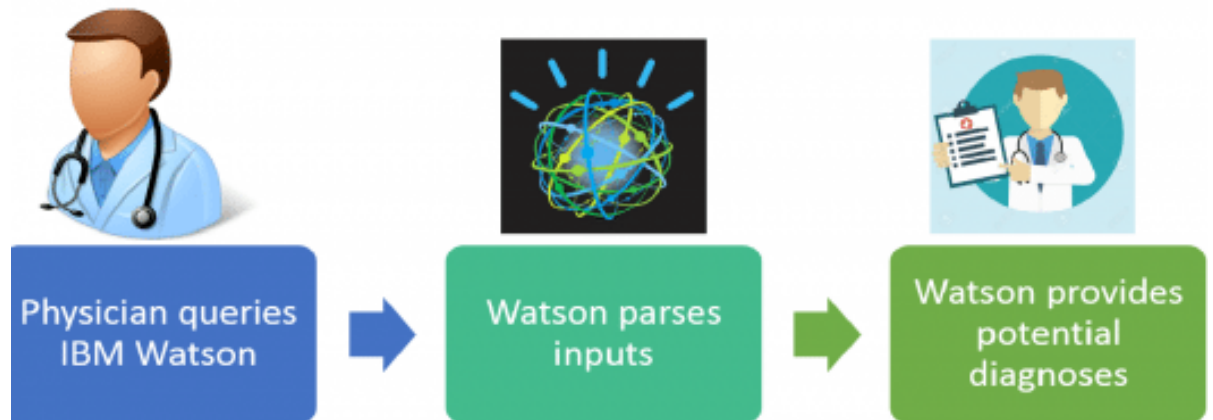
Big data analytics

- Big data is **more real-time in nature** than traditional data warehouse applications.
 - Conventional architectures are ill-suited for big data apps (e.g., Exadata, Teradata)
- There are many challenges in handling Big data, mainly about the **technology bottleneck** and the **lack of experts**.

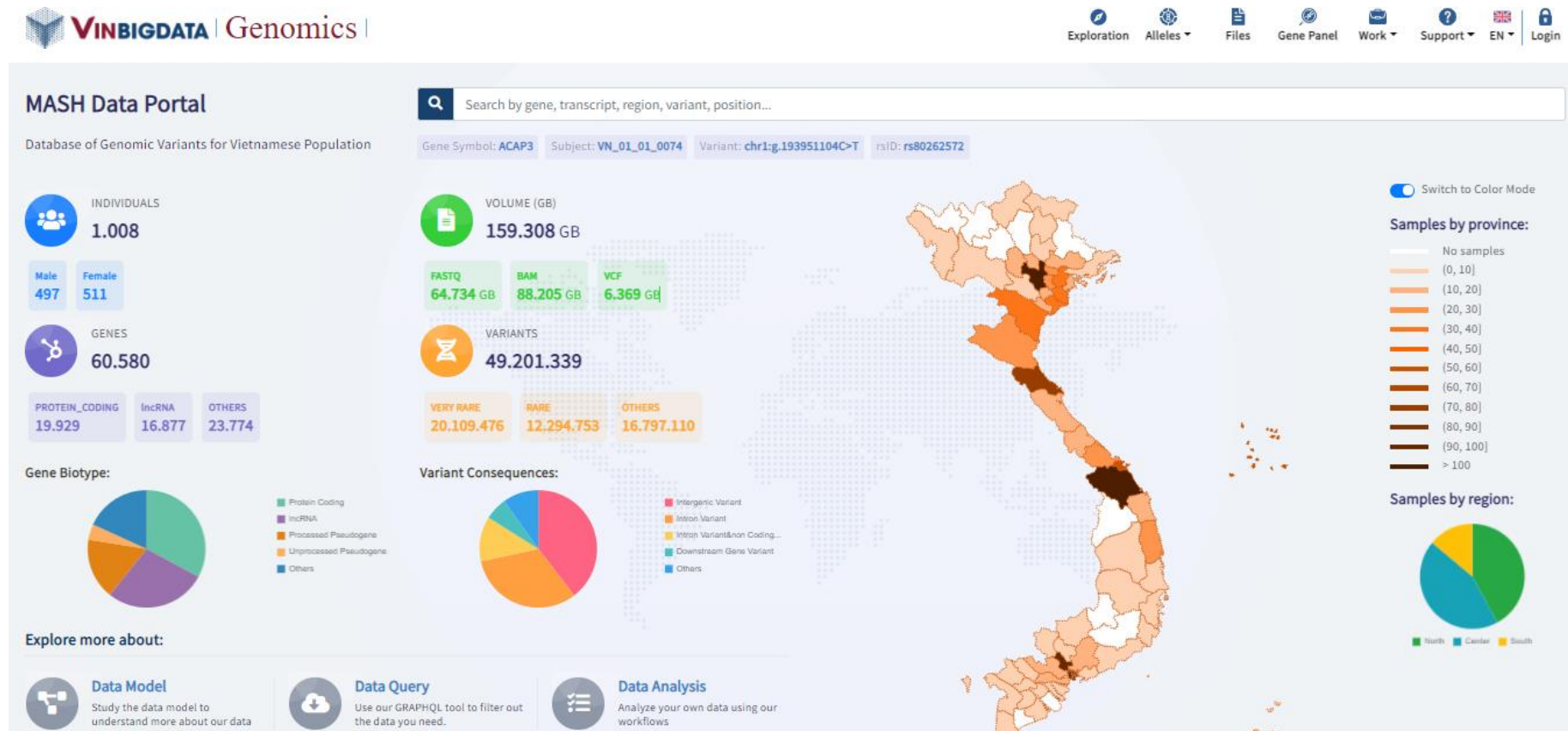


Big data in Healthcare

- 80% of medical data is unstructured and clinically relevant, residing in multiple places.
 - Individual EMRs, labs and imaging systems, physician notes, medical correspondence, etc.
- Big data may help increase access to healthcare.
 - Build sustainable healthcare systems, collaborate to improve care and outcomes.



Big data in Healthcare



VinBigData's Genomics project.
Homepage: <https://genome.vinbigdata.org/>

KTH: Reducing traffic congestion



KTH Swedish Royal Institute of Technology Reducing Traffic Congestion

Capabilities Utilized:

Stream Computing

- Deployed real-time Smarter Traffic system to predict and improve traffic flow.
- Analyzes streaming real-time data gathered from cameras at entry/exit to city, GPS data from taxis and trucks, and weather information.
- Predicts best time and method to travel such as when to leave to catch a flight at the airport

Significant benefits:

- Enables ability to analyze and predict traffic faster and more accurately than ever before
- Provides new insight into mechanisms that affect a complex traffic system
- Smarter, more efficient, and more environmentally friendly traffic



SONAR:
~10-100 KB
per second

RADAR:
~10-100 KB
per second

GPS:
~50 KB
per second

CAMERAS:
~20-40 KB
per second

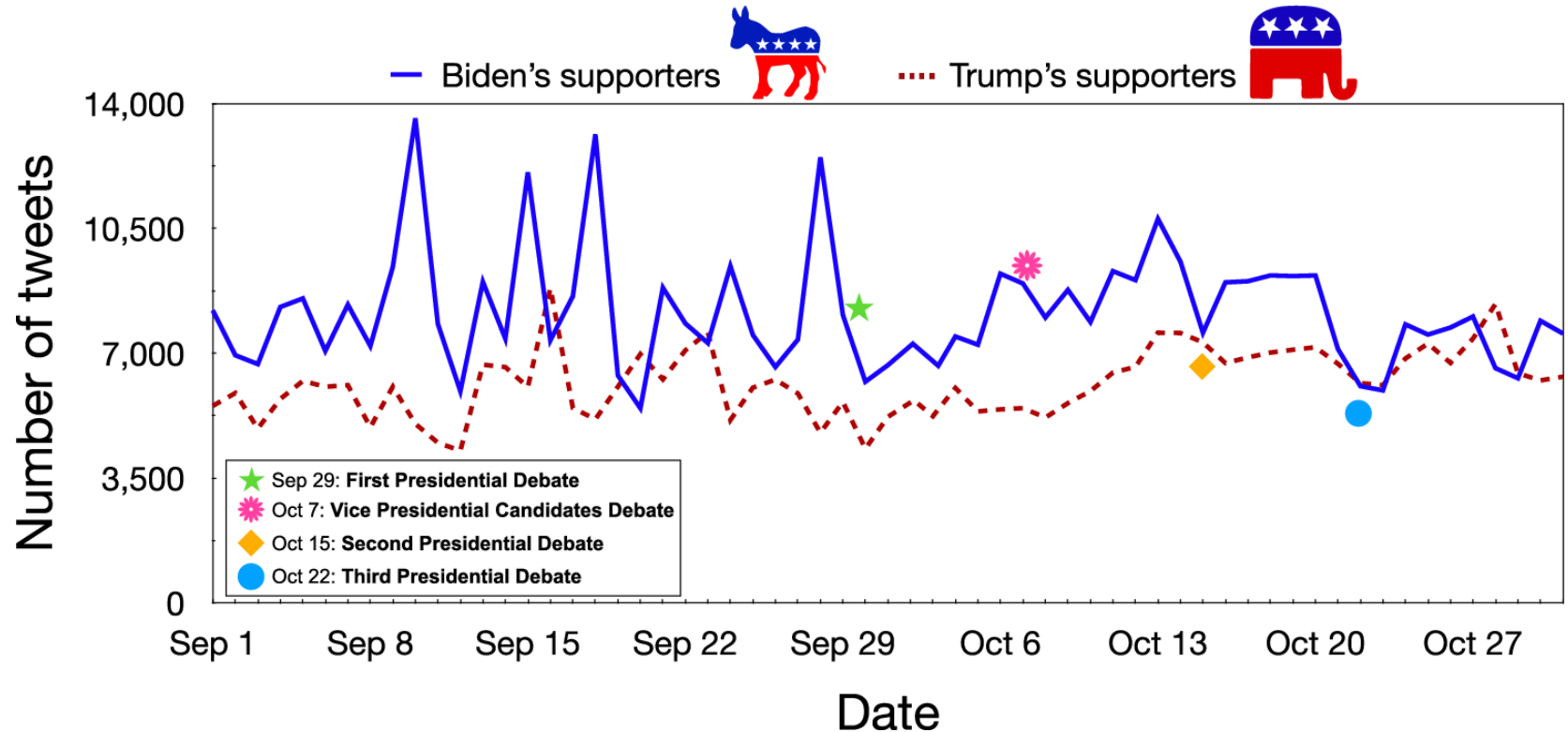
LIDAR:
~10-70 KB
per second

AUTONOMOUS VEHICLES GET

4,000 GB

DATA EACH DAY

Big data in social network analysis



Time series of polarized tweets published from September 1 to October 31, 2020.

Belcastro, L., Branda, F., Cantini, R. et al. Analyzing voter behavior on social media during the 2020 US presidential election campaign. Soc. Netw. Anal. Min. 12, 83 (2022).

Motivations and opportunities

A new horizon that changes our lives...

New insight into data

- Why deal with more data? *New insights.*
- The insights are for people throughout the enterprise, not top-level executives only.
 - People involved may include the CEO, marketing staffs, data analyst and programmers.
- They can be transformed into actionable intelligence
 - better business, better service to customers
 - more profit gained

Some real-world applications



310 million active users globally



12 million items across all its categories and services

Over 1 billion monthly active users



There were a total of 8.6 billion videos uploaded in TikTok in 2021. The number in 2023 approximates 14.4 billion videos.



STEAM®

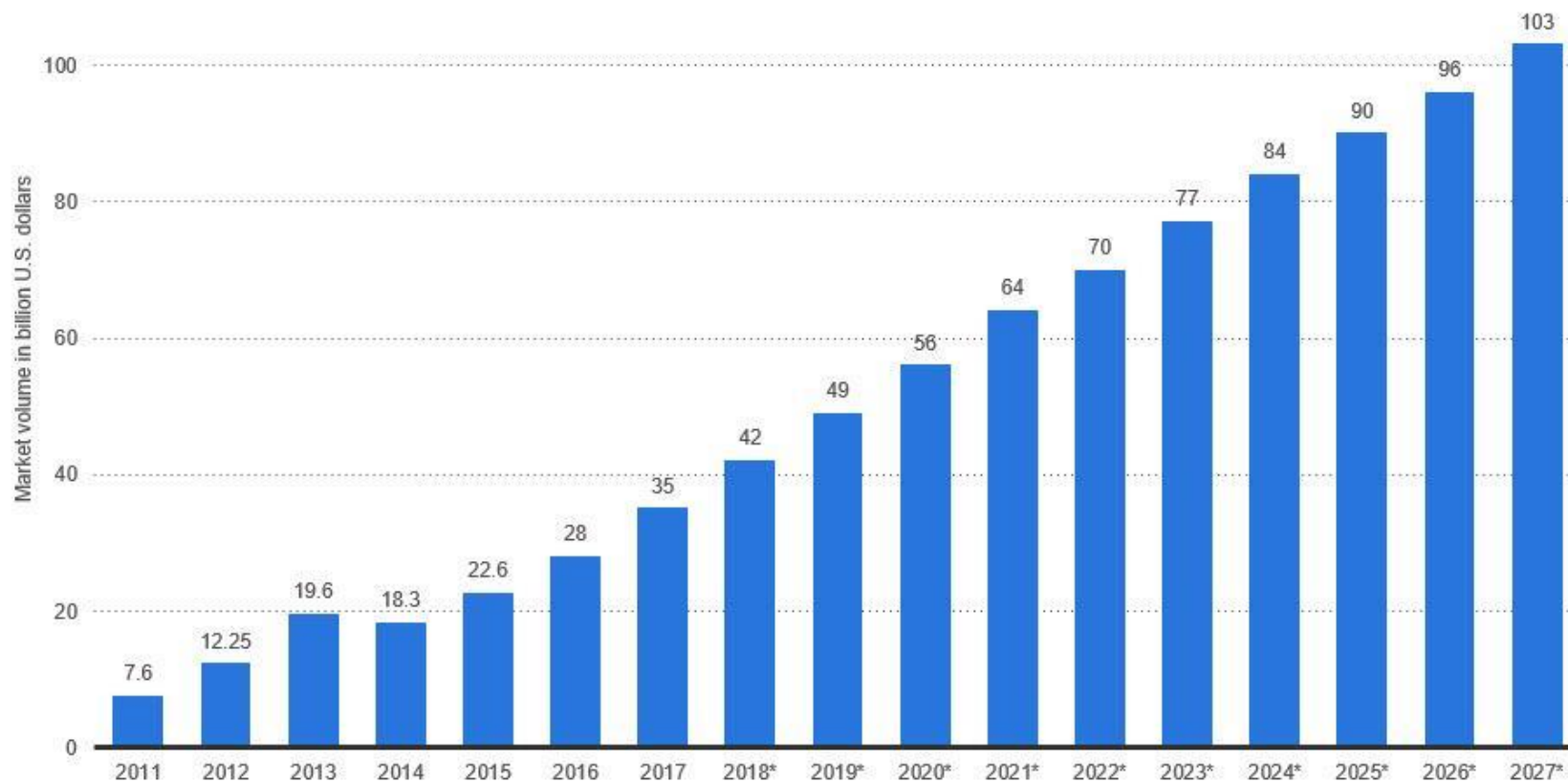


20,919,374 players online, 5,163,503 players in-game

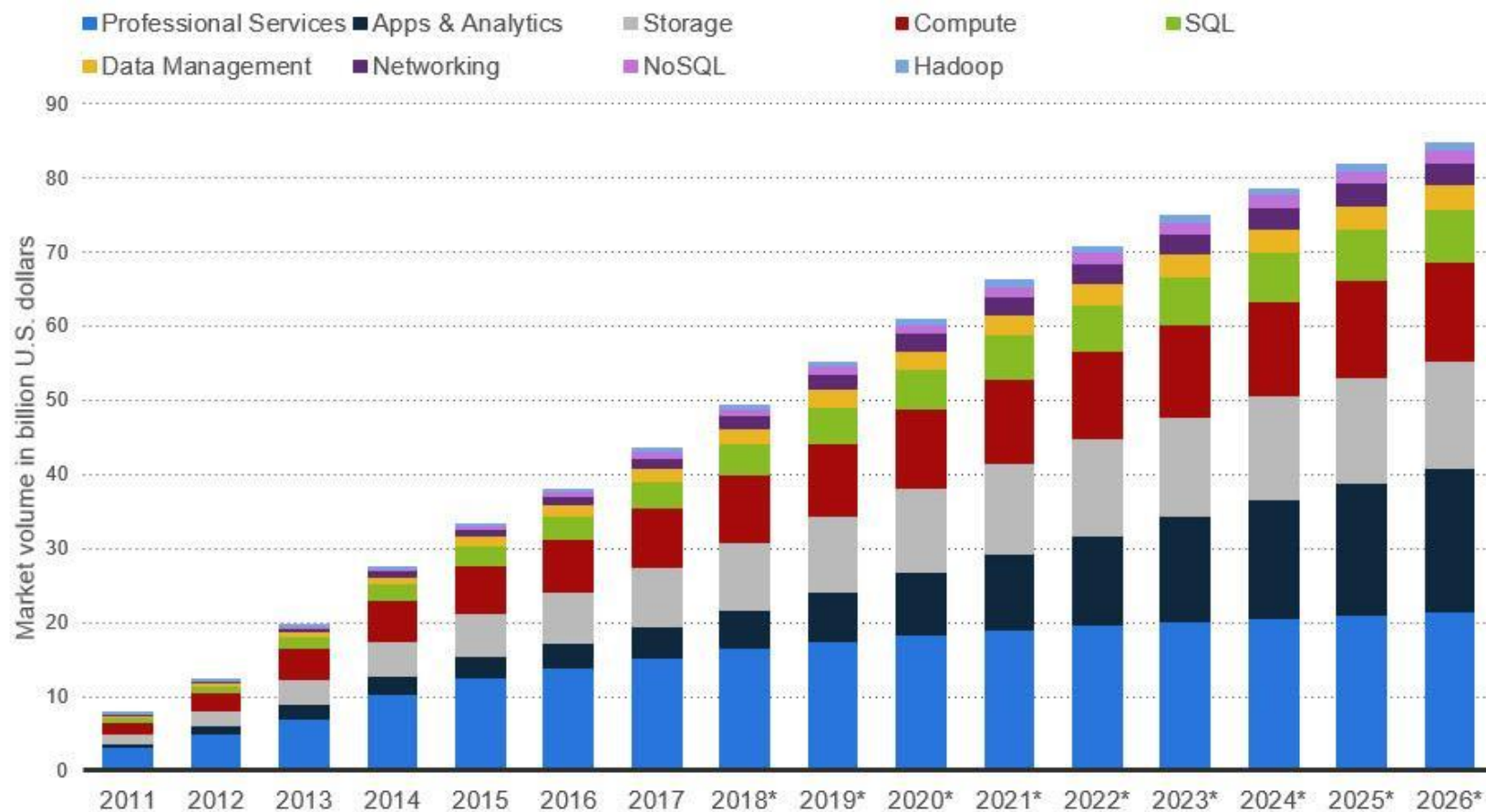


105,281 tracked Steam games

Big Data Market Size Revenue Forecast Worldwide From 2011 To 2027 (in billion U.S. dollars)

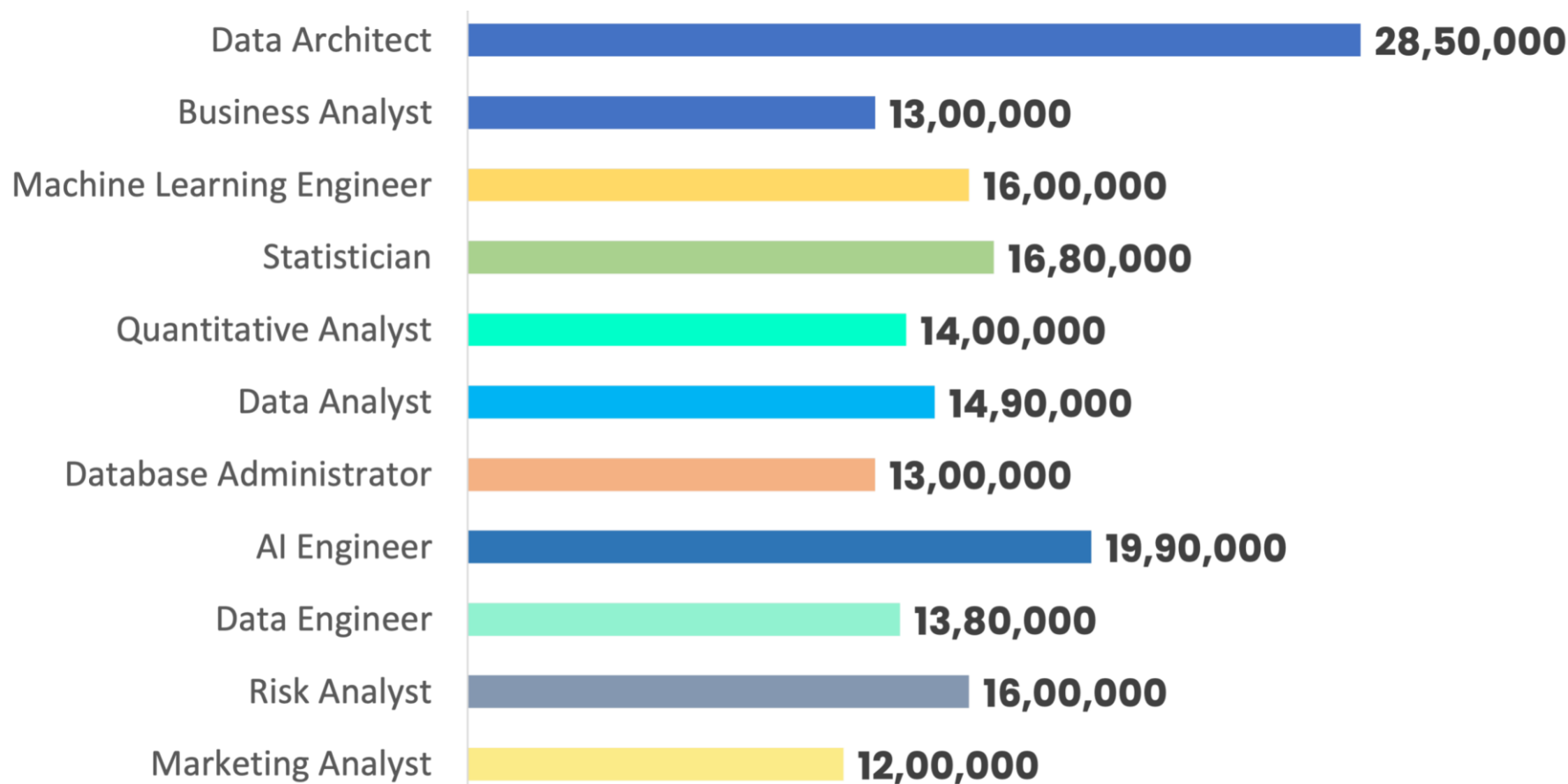


Big Data Market Forecast Worldwide from 2011 to 2026, by segment (in billion U.S. dollars)



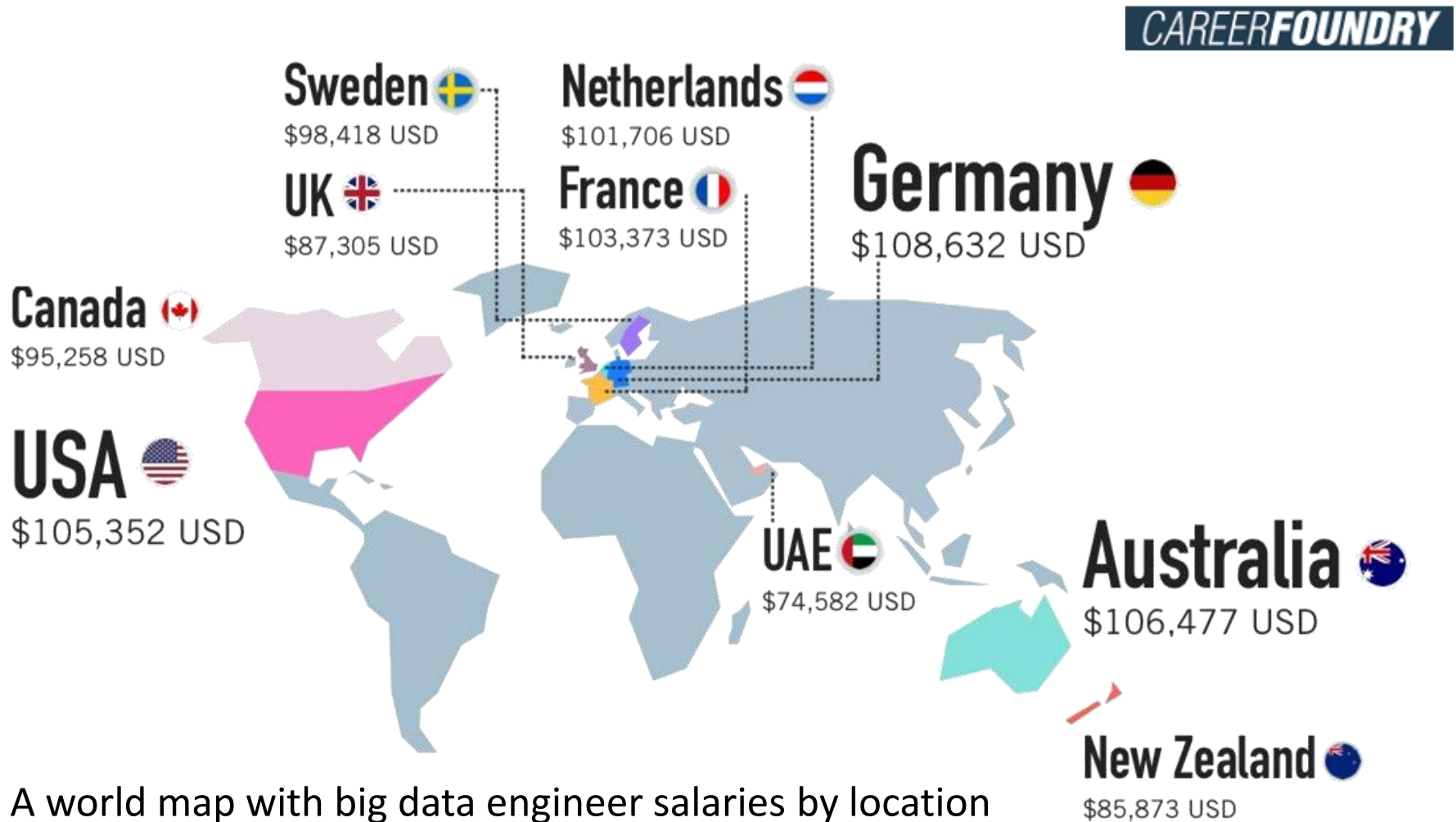
Salary by Job roles in India (July 2023)

ANALYTIX LABS



* The currency unit is INR (Indian Rupee)

Big data engineer salaries by location



A world map with big data engineer salaries by location
(updated in January 2023)

Average big data engineer salary

CAREERFOUNDRY

(updated in January 2023)

 **Meta**

\$229K



\$205K

 **Microsoft**

\$183K

 **amazon**

\$167K

 **Apple**

\$170K

...the end.

